

Portfolio Construction: Risk Allocation v. Return Optimization

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Abstract

This paper compares two major portfolio construction frameworks: mean-variance optimization (MVO), which allocates capital by a trade-off between expected return and variance, and risk parity, which allocates capital by equalizing risk contributions across assets. The analysis combines theory with an empirical out-of-sample comparison using a common seven-asset ETF universe: ACWI, TLT, LQD, TIP, DBC, VNQ, and QAI. Portfolios are estimated with a rolling 252-trading-day window, rebalanced monthly under long-only, fully invested constraints. Using closing-price based arithmetic returns from April 2016 through April 2026, I evaluate annualized return, volatility, Sharpe ratio, drawdown, turnover, diversification, and sensitivity to estimation error. In this sample, MVO delivers higher annualized return and Sharpe ratio, but it is also more concentrated, less stable, and more sensitive to estimation error. Risk parity delivers lower raw performance, but it exhibits lower volatility, smaller drawdowns, higher diversification, and lower turnover. These findings illustrate the practical balancing between return optimization and risk allocation.

I. Introduction

Portfolio construction lies at the center of investment management because the same asset universe can produce very different investor experiences depending on how weights are chosen. The classic framework is Markowitz mean-variance optimization (1952), which selects portfolio weights to maximize expected return relative to risk. If expected returns and covariances are known, MVO identifies an efficient allocation. In practice, however, these inputs are estimated with error, and small changes in the inputs can lead to large changes in the portfolio.

On the other hand, risk parity approaches the problem by determining how portfolio risk should be distributed across assets. In the equal risk contribution version (Asness et al., 2012; Maillard et al., 2010), each asset is assigned the same contribution to total portfolio volatility. The model relies solely on the covariance matrix, eliminates direct dependence on expected-return estimates, and often leads to more diversified allocations. Thus, risk parity is often viewed as more robust in realistic settings where estimates are noisy and unstable.

The purpose of this paper is to compare these two models on both theoretical and empirical grounds. The theoretical comparison highlights the role of expected returns, covariance estimation, and diversification. The empirical section then examines how the two methods perform in a realistic exercise when parameters are estimated through time, and weights are applied only to future returns. This out-of-sample comparison helps avoid look-ahead bias.

II. Theoretical Framework

A. Mean-Variance Optimization

MVO chooses a weight vector w to maximize expected portfolio utility, commonly written as:

$$\max[E(R_p) - \frac{\lambda}{2} \text{Var}(R_p)],$$

where $E(R_p) = w^T \mu$ and $\text{Var}(R_p) = w^T \Sigma w$.

In the unconstrained case, the optimal solution is proportional to $\Sigma^{-1} \mu$. The optimizer depends heavily on the estimated mean vector μ , which is extremely unstable in finite samples. Once realistic constraints such as long-only weights and full investment are added, the optimization must be solved numerically, and the resulting portfolio can still be highly concentrated if one or two assets appear temporarily attractive relative to the rest.

The strength of MVO is that it is grounded in expected utility theory and explicitly incorporates return expectations. Its weakness is that it can amplify estimation noise, particularly in the mean-return estimates. In practice, this often leads to unstable and corner-solution portfolios.

B. Risk Parity (Equal Risk Contribution)

Risk parity starts the portfolio construction by equal weight allocation so that each asset contributes equally to total portfolio risk. If portfolio volatility is $\sigma_p = \sqrt{w^T \Sigma w}$, then the marginal contribution of asset i to volatility is $\frac{(\Sigma w)_i}{\sigma_p}$, and its total risk contribution is:

$$RC_i = w_i \frac{(\Sigma w)_i}{\sigma_p}.$$

Equal risk contribution requires $RC_i = RC_j$ for all assets i and j . In practice, this is solved numerically subject to the portfolio constraints. The main strength of risk parity is that as it depends primarily on the covariance matrix, it is typically less sensitive to noise in the data. The

trade-off is that it may tilt heavily toward lower-volatility assets, generating lower expected returns. Institutional investors may use leverage to scale the whole portfolio to a higher target risk level. In this paper, both portfolios are constrained to be long-only and fully invested, so gross leverage always equals to one by construction.

III. Data and Methodology

A. Asset Universe

I use a diversified seven-asset ETF universe to represent major macro risk exposures: international equities (ACWI), long-duration Treasury bonds (TLT), investment-grade credit (LQD), inflation-linked bonds (TIP), commodities (DBC), real estate (VNQ), and alternatives (QAI). This set minimizes overlap from combining broad equity funds with narrower U.S. equity index funds like VOO and provides a clean comparison between return-seeking allocations and risk-balanced allocations.

B. Data

The data consist of daily closing prices from April 2016 through April 2026, and returns are computed as arithmetic percentage returns. Because adjusted prices were unavailable, measured returns likely understate the true performance of income-distributing ETFs such as TLT, LQD, TIP, VNQ, and QAI. This limitation should be kept in mind when interpreting the level of returns, although the comparison remains informative.

C. Portfolio Constraints

Both strategies are implemented under long-only, fully invested constraints. This keeps the exercise investable and comparable across methods, avoids additional assumptions required for short-selling, and limits the extreme weight patterns that unconstrained MVO can generate.

Under these constraints, the sum of portfolio weights is one.

D. Rolling Estimation and Rebalancing

The empirical design uses a rolling 252-trading-day estimation window, roughly one year of daily data. At each month-end rebalance date, I estimate the required inputs using only the previous 252 trading days. The resulting weights are then held during the following month. At the next month-end, the window rolls forward and the process repeats. This rolling structure is designed to mimic information exposure to an investor in reality and prevents look-ahead bias.

E. Performance and Stability Metrics

Out-of-sample performance is evaluated using annualized return, annualized volatility, Sharpe ratio, and maximum drawdown. To assess implementation practicality, I also compute average monthly turnover, and effective number of assets. The effective number of assets is defined as the equivalent number of equally weighted assets in a portfolio, calculated as the reciprocal of the sum of squared weights of the portfolio's assets. Higher values indicate greater diversification, while lower values indicate a more concentrated portfolio. Sensitivity to estimation error is studied through bootstrap resampling of each rolling estimation window. For each rebalance date, I compare the baseline weights to weights obtained from resampled windows using the L1 distance between weight vectors. Parameter instability is summarized by the variability of rolling annualized mean and volatility estimates across assets.

IV. Empirical Results

The out-of-sample results reveal a clear trade-off between higher realized return and greater robustness. Table 1 summarizes the main performance statistics.

Table 1: Performance Statistics of MVO and Risk Parity

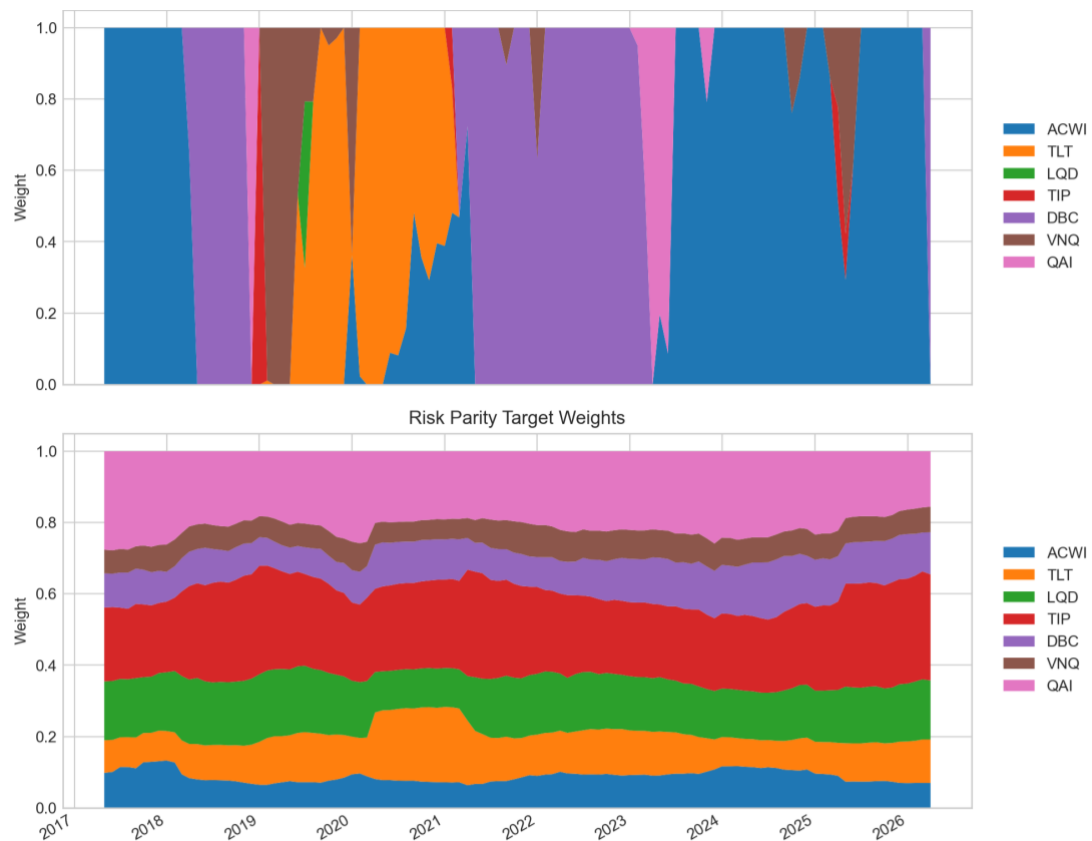
Metrics	MVO	Risk Parity
Annual return	8.15%	1.36%
Volatility	16.31%	7.00%
Sharpe ratio	0.50	0.19
Max drawdown	-26.23%	-20.42%
Average monthly turnover	16.03%	2.05%
Effective number of assets	1.20	5.81

In this sample, the MVO portfolio delivers higher annualized return and higher Sharpe ratio (Figure 1), also accompanied by more volatility and a very small number of positions. The effective number of assets for MVO is only 1.2 on average, which means the optimizer is often behaving like a near single-asset portfolio. This pattern is visible in the portfolio weights, where MVO frequently moves toward corner solutions (Figure 2). Such concentration is a classic practical criticism of MVO.

Figure 1: Out-of-Sample Cumulative Returns of MVO and Risk Parity



Figure 2: Target Weights of MVO and Risk Parity



In comparison, the risk parity portfolio produces a lower annualized return, but its volatility is significantly less than that of MVO, and the maximum drawdown is smaller. The portfolio is also genuinely diversified, with an average effective number of assets of 5.81. This reflects the central goal of equal risk contribution: distributing portfolio risk broadly rather than letting one asset dominate the portfolio.

V. Sensitivity to Estimation Error and Parameter Instability

A central limitation of MVO is its high sensitivity to estimation error. The results here strongly support that claim. Using bootstrap resampling of the rolling estimation windows, the mean L1 change in MVO weights is 94.62%, compared with only 5.94% for risk parity. At the 90th percentile, the MVO weight change reaches 200.00%, compared with only 9.90% for risk parity.

These numbers show that small shifts in the estimation sample can dramatically alter the MVO portfolio. Intuitively, this occurs because the optimizer reacts aggressively to differences in estimated expected returns, even when those differences are mostly noise. Risk parity is not immune to estimation error, since the covariance matrix is estimated from the same data, but it is far less fragile because it does not rely directly on expected returns, the most unstable input.

The results also show that estimated expected returns vary more than estimated volatilities over time. Across assets, the average standard deviation of rolling annualized mean estimates is 11.49%, while that of annualized volatility estimates is only 4.37%. As MVO depends heavily on expected return estimates, it is highly sensitive to this instability. Risk parity relies primarily on volatility and covariance estimates, which are generally more stable and make the portfolio less sensitive to estimation noises.

VI. Practical Considerations

A. Turnover

Turnover is a practical measure of how much trading a strategy requires. MVO has average monthly turnover of 16.03%, compared with only 2.05% for risk parity. Higher turnover implies greater transaction costs, more implementation drag, and less stability for the investor. This is an important practical disadvantage of MVO even before considering taxes or market impact.

B. Diversification

The effective number of assets provides a compact measure of diversification. Risk parity's value of 5.81 indicates broad participation by most assets in the portfolio, whereas MVO's value of 1.20 indicates higher concentration. In other words, even under long-only constraints, the return-optimization approach tends to produce portfolios that are dominated by a small subset of the opportunity set.

VII. Discussion and Conclusion

The results of this study suggest that neither portfolio construction framework is universally superior; rather, each reflects a different investment philosophy and set of practical trade-offs. MVO is attractive when expected-return estimates are sufficiently reliable and the investor is willing to accept concentration in pursuit of higher expected payoff. In the present sample, MVO achieved the higher annualized return and Sharpe ratio, but it did so at the cost of higher volatility, deeper drawdowns, higher turnover, and lower diversification. Risk parity, by contrast, delivered lower raw performance, but it produced a more balanced and stable portfolio profile.

From a practical portfolio-management perspective, the most important result may be the contrast in robustness. The MVO portfolio proved highly sensitive to small changes in the estimation sample and frequently moved toward corner solutions, while risk parity remained diversified and comparatively stable over time. This finding is consistent with the portfolio construction literature: the main challenge of MVO is the instability of the expected-return inputs on which it depends. Risk parity sacrifices some return-seeking flexibility yet gains resilience when estimation error is unavoidable.

Overall, the comparison highlights the fundamental trade-off between return optimization and risk allocation. Investors who prioritize higher expected payoff and are comfortable with model risk may prefer MVO, especially if expected-return estimates can be improved. Investors who place greater value on diversification, lower turnover, and implementation robustness may prefer risk parity. Future research could extend this analysis by incorporating shrinkage estimators, Black-Litterman expected returns, transaction-cost penalties, or volatility-targeted leverage for risk parity.

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